p61 selected-degree adjacency

Replayed production calibration and 6 TB corpus contract

Field	`p = 2^61 - 1 = 2305843009213693951`
Selected class	`p61-i0025`, trace `-704859453`
Class order	`358094304` vertices
Frozen degrees	`5, 13, 37, 59`
Run	`p61-adjacency-pilot-20261015T0436Z`

PRODUCER PASS INDEPENDENT REPLAY PASS FULL GRAPH ACTIVE / PARTIAL

Decision

A 4,096-vertex production window is admitted with 28,672 directed edges, 471,040 kernel coefficients, zero generation failures, and exact kernel-first replay. The calibrated representation projects to 3.632 TB logical JSONL and 0.748 TB gzip for the complete selected graph. Coverage is still 4,096 of 358,094,304 source vertices.

Producer `f9b48fcf`; verifier `92a62be6`; different source commits, binaries, and reconstruction paths.

1. Evidence state

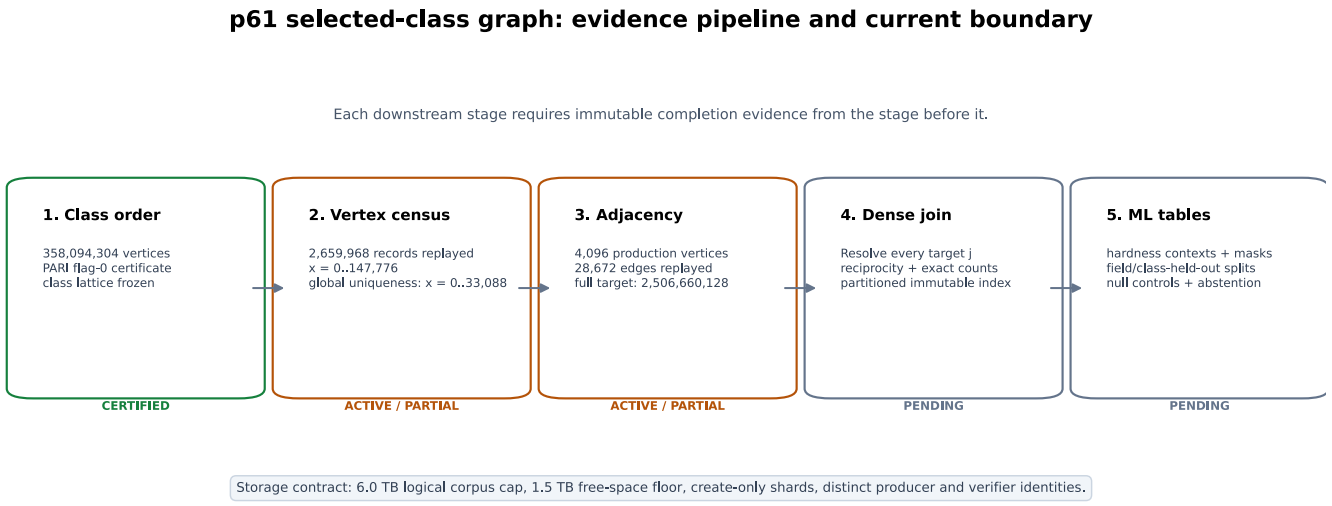


Figure 1: Downstream graph and ML products remain gated on complete upstream evidence.

Requirement	State	Boundary
Class size	**Certified**	358,094,304 vertices; receipt bound into every shard
Four-degree edge path	**Pass**	1/2/2/2 rational edges per vertex on every admitted window
Independent replay	**Pass**	Kernel-first verifier at a distinct commit and binary digest
Vertex census	**Partial**	2,659,968 records replayed through `x=147776`
Global uniqueness	**Partial**	595,584 records externally deduplicated through `x=33088`
Production adjacency	**Partial**	4,096 / 358,094,304 source vertices
Dense join + reciprocity	Pending	Requires the complete globally unique vertex index
ML-ready corpus	Pending	Labels, masks, split map, null controls, and joined targets remain

2. Frozen graph arithmetic

Degree	Behavior	Neighbors	Full edges
5	ramified	1	358,094,304
13	split	2	716,188,608
37	split	2	716,188,608
59	split	2	716,188,608
Total		**7**	**2,506,660,128**

The complete selected graph also contains exactly 41,180,844,960 kernel coefficients. Other degrees, trace classes, and the field-wide curve population are separate scopes.

3. Replayed calibration

Window	Vertices	Edges	Generate	Replay	Logical	Gzip
Path smoke	1	7	18.703 s	21.446 s	12,227	3,593
Admission	64	448	24.347 s	23.712 s	649,507	135,394
Production	4,096	28,672	191.551 s	296.931 s	41,543,871	8,552,556

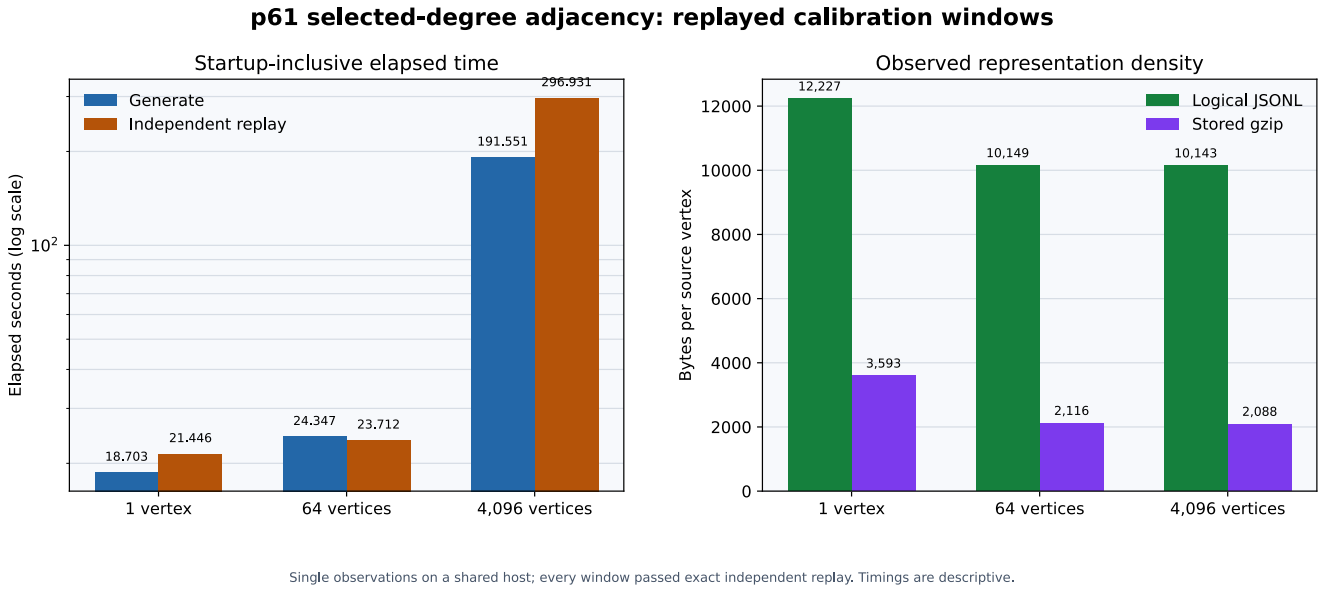


Figure 2: Single startup-inclusive observations on a shared host. Every window passed exact independent replay.

The production window recorded 21.383 generation vertices/s, 13.794 replay vertices/s, and 8.385 vertices/s for the sequential admitted pipeline. A linear single-worker diagnostic is 193.8 generation days plus 300.5 replay days. It is not a schedule commitment: the eight-thread vertex census was active and no isolation or repeated timing design was used.

4. Storage contract

Projected product	Bytes	TB
Complete adjacency JSONL	3,631,988,176,565	3.632
Complete adjacency gzip	747,710,348,692	0.748
Logical allowance after adjacency	2,368,011,823,435	2.368

The remaining allowance is reserved for useful dense joins, reciprocity audits, frozen attack contexts, observation masks, split maps, and ML-ready tables. Logical and compressed sizes remain separate. The supervisor enforces a 1.5 TB free-space floor and a 6.0 TB logical cap.

5. Completion boundary

The vertex stream has per-segment replay through `x=147776`, but global uniqueness is sealed only through `x=33088`. The adjacency stream therefore records canonical target j-invariants without assigning dense target indices. The early `x=0..16704` prefix also requires a normalized source catalog or bounded backfill because it predates the campaign-admission schema.

Terminal completion requires:

1. every one of 358,094,304 sources exactly once;
2. exactly 2,506,660,128 directed edges and the fixed per-degree totals;
3. a globally unique dense vertex index with every target resolved;
4. reciprocal endpoint and degree consistency;
5. immutable manifests for every shard, retry, join partition, and exception; and
6. frozen hardness labels, explicit censor masks, field/class split maps, abstention, and null controls before M7 successor training.

6. Preserved failures and limitations

- An exact-producer reconstruction under `/private/tmp` exhausted local scratch during archive extraction. The corrected detached worktree on SSD990-2 built in 602 seconds and reproduced the pilot binary digest exactly.
- The first supervisor unit-test invocation failed during dynamic module registration. The corrected final suite passes five tests, including the bounded-follow guard.
- SMB per-file sync is unsupported and remains recorded. Host sync and post-write digest replay precede admission.
- Timings are descriptive shared-host observations, not a comparative speed result.

7. Evidence index

Artifact	SHA-256
Producer binary	`a69def0c21dc455077abb8bef86d828c13dfcbea8f45382f71614fdf260d7c78`
Verifier binary	`85b1265b187ecd2bbd2ff7fd0f7c9608d3ccd2e4a7234f73ee1b2863be599208`
64-vertex admission	`26b4358961f6cf3b58767b454a5272b101ea520a14ebb95ad409ded7bc37685f`
4,096-vertex shard	`ce0fd43d317b56e386462b7cb8ebd5605bba1195efaa57ba1e36819c5917bbc1`
4,096 generation receipt	`81fc61cf45c2718acd74236ebff5201f1e751c570c57afc86cdd1491a0526cbf`
4,096 replay receipt	`5755b14028bcf7bf51e19870ae3fa8e6422661e9836a6fc8780ead7e8878870e`
4,096 admission	`be33159cc6fb7bc3460c6626c827d2e01ea2cf46ada017fb8acafc19b21e26e8`

Durable root

`/Volumes/data/ec-hardness-atlas-20261012/p61-selected-adjacency-v1/`